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## LangSmith Self-Hosted | Production AWS/EKS Reference

**Corporate users + agent SDKs**

Via VPN or Direct Connect
Private LangSmith access

Amazon Route 53

AWS Cloud

AWS Region eu-west-1 — design assumption

VPC 10.40.0.0/16 — illustrative addressing

Public subnet • AZ A

**NAT Gateway A**
AZ-local private outbound

NAT gateway

Public subnet • AZ B

**NAT Gateway B**
AZ-local private outbound

NAT gateway

Public subnet • AZ C

**NAT Gateway C**
AZ-local private outbound

NAT gateway

**Internal Application Load Balancer**
TLS + multi-AZ frontend targets

Application Load Balancer

ONE EKS CLUSTER — WORKER COMPUTE SPANS 3 AVAILABILITY ZONES

EKS

**Amazon EKS managed control plane**
AWS runs control plane; our EC2 workers host the actual pods.

**Namespace: langsmith (logical)**
Deployments are cluster-wide and replicas can land in any AZ.

**Pod HPA / KEDA + EC2 node autoscaling**
Separately scale pods and worker node capacity.

AZ A (placement view)

Private EKS worker subnet

EKS managed node group • EC2 Auto Scaling

EC2 Worker 1 (AZ A)

EC2

kubelet · runtime
Kubernetes node

**POD — Frontend**
Serves the LangSmith web UI through Nginx and routes incoming API requests to the appropriate backend services.

**POD — Backend**
Handles core business logic, including projects, datasets, experiments, configuration, and trace queries.

**POD — Platform Backend**
Handles authentication and high-volume trace ingestion, separating ingestion traffic from regular API operations.

**POD — Playground**
Forwards prompt execution requests to configured LLM providers and model endpoints.

EC2 Worker 2 (AZ A)

EC2

kubelet · runtime
Kubernetes node

**POD — Queue**
Processes asynchronous background jobs, including trace and feedback processing, validation, and retries.

**POD — Ingest-queue**
Consumes buffered trace data from Redis and asynchronously persists it to ClickHouse.

**POD — ACE Backend**
isolated custom code

**Scheduling headroom**
Executes custom evaluator code and other supported operations in an isolated execution environment.

AZ B (placement view)

Private EKS worker subnet

EKS managed node group • EC2 Auto Scaling

EC2 Worker 1 (AZ B)

EC2

kubelet · runtime
Kubernetes node

**POD — Frontend**
Nginx; web + API ingress

**POD — Backend**
CRUD; trace query path

**POD — Platform Backend**
authentication + trace intake

**POD — Playground**
LLM API forwarding

EC2 Worker 2 (AZ B)

EC2

kubelet · runtime
Kubernetes node

**POD — Queue**
async processing / retries

**POD — Ingest-queue**
Redis → ClickHouse persistence

**POD — ACE Backend**
isolated custom code

**Scheduling headroom**
Additional pod replicas can be scheduled.

AZ C (placement view)

Private EKS worker subnet

EKS managed node group • EC2 Auto Scaling

EC2 Worker 1 (AZ C)

EC2

kubelet · runtime
Kubernetes node

**POD — Frontend**
Nginx; web + API ingress

**POD — Backend**
CRUD; trace query path

**POD — Platform Backend**
authentication + trace intake

**POD — Playground**
LLM API forwarding

EC2 Worker 2 (AZ C)

EC2

kubelet · runtime
Kubernetes node

**POD — Queue**
async processing / retries

**POD — Ingest-queue**
Redis → ClickHouse persistence

**POD — ACE Backend**
isolated custom code

**Scheduling headroom**
Additional pod replicas can be scheduled.

Private database subnet • AZ A

Amazon RDS

**Transactional database (OLTP)**

Stores LangSmith's operational data, including organizations, users, projects, API keys, datasets, and platform configuration.

Amazon RDS provides managed PostgreSQL with automated backups, patching, encryption, and Multi-AZ failover.

**Primary database**

Amazon ElastiCache

**In-memory queue and cache**

Provides fast in-memory storage for LangSmith's asynchronous trace and feedback ingestion, background job queues, and caching.

Buffers incoming telemetry before asynchronous workers persist it to ClickHouse; decoupling ingestion from database write performance.

**Primary instance**

Private database subnet • AZ B

Amazon RDS

**Transactional database (OLTP)**

**Secondary database**

Amazon ElastiCache

**In-memory queue and cache**

**Secondary instance**

Private database subnet • AZ C

**RDS Multi-AZ subnet group**
Zone can participate in DB failover design

Amazon ElastiCache

**In-memory queue and cache**

**Third instance**

Amazon Simple Storage Service (Amazon S3)

**Amazon S3**
Blob artifacts + attachments

Amazon CloudWatch

**CloudWatch**
metrics / logging / alarms

AWS Secrets Manager

**Secrets Manager**
keys / database credentials

AWS Key Management Service (AWS KMS)

**AWS KMS**
Encryption key control

**External ClickHouse cluster — OLAP**
Trace spans and feedback. Scales independently from EKS; choose private peering or supported PrivateLink. Not an AWS-managed ClickHouse icon.

**1 / NETWORK AND INGRESS**
Private clients use corporate access and Route 53 private DNS. Internal ALB sends traffic to the Frontend Kubernetes service across AZs. NAT gateways in public subnets supply controlled outbound connectivity.

**2 / EKS, SUBNETS, WORKER NODES**
Amazon EKS is one managed Kubernetes cluster. Its private worker nodes run as EC2 instances in three AZs. Node groups provision capacity. Each EC2 worker has kubelet and may host multiple application pods.

**3 / NAMESPACE → DEPLOYMENTS → PODS**
The namespace and workloads are logical resources across the cluster, not physically contained in one AZ. This page shows illustrative scheduling of all seven workloads: Frontend, Backend, Platform Backend, Playground, Queue, Ingest-queue, ACE Backend.

**4 / STATEFUL SERVICES AND HA**
PostgreSQL RDS + ElastiCache Redis/Valkey are outside EKS in private data subnets; ClickHouse is external; S3 is regional. Multi-AZ placement and replication are design choices, requiring recovery and failover testing.

**5 / INDEPENDENT SCALING**
HPA scales Frontend/Backend/Platform Backend by CPU or memory. KEDA scales Queue and Ingest-queue using backlog (v0.13+). EC2 node autoscaling provides capacity for pending pods. PDBs and topology spread improve resilience.

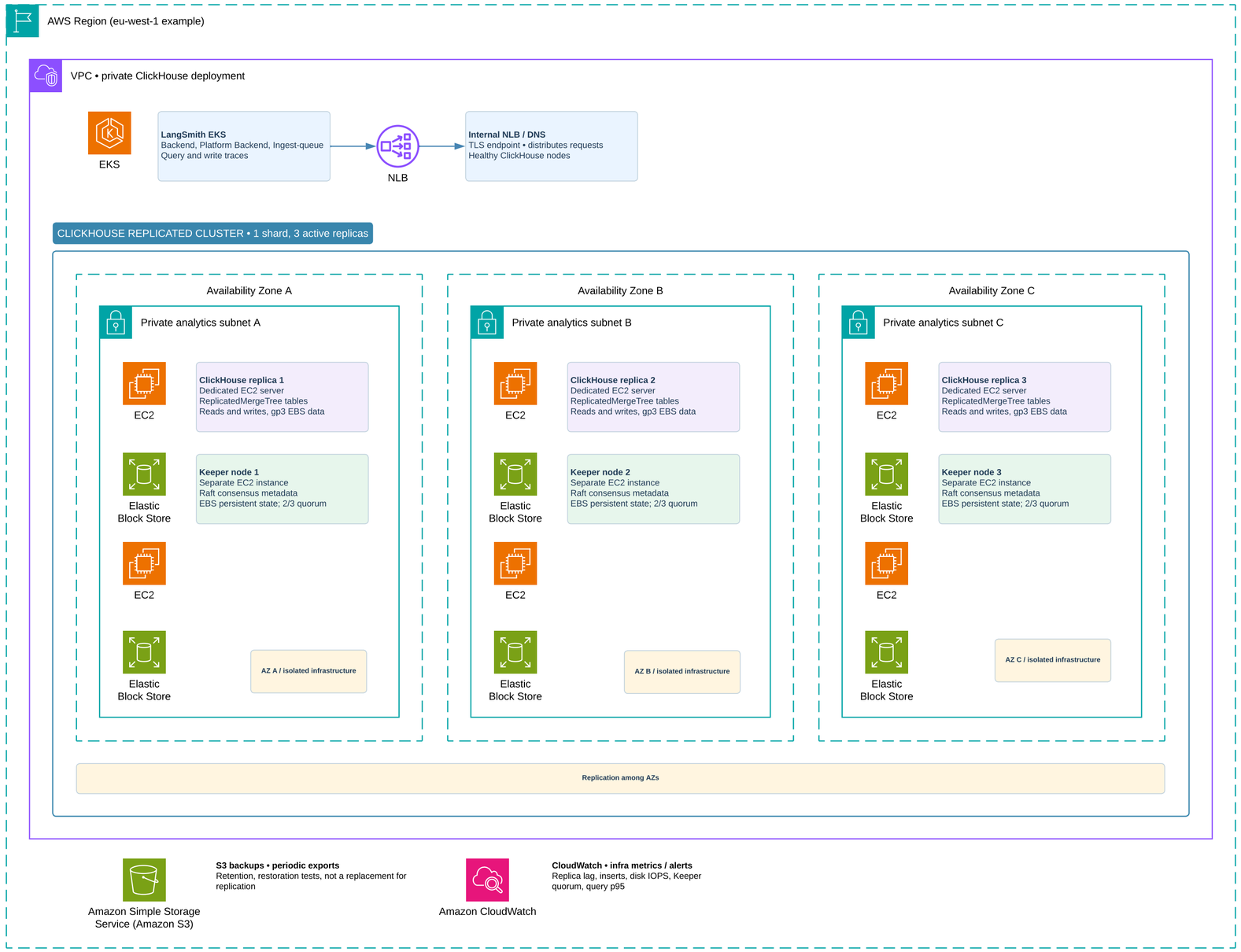
**6 / RISKS AND CUSTOMER DECISIONS**
Resolve trace volume, burst patterns, query concurrency, data TTL, RPO/RTO, identity provider, model egress, ClickHouse network path and data residency. The 3-AZ architecture is a proposal, not a LangSmith hard requirement.

**IMPORTANT SEMANTIC NOTE: NESTING IS FOR READABILITY**
AWS Availability Zones exist outside EKS; the EKS cluster logically spans these AZs. The namespace spans all nodes. Pod placements and primary/standby labels are illustrative, not guaranteed replicas or live deployments.

**ARCHITECTURAL BASELINE & OPEN QUESTIONS**
Official Kubernetes baseline is 16 vCPU / 64 GB available (not a capacity guarantee). Confirm peak traces/sec and size, read concurrency, retention, zone failure, security, SSO, costs, available storage capabilities, operating SLO and disaster recovery.

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## ClickHouse Self-Hosted | 1 shard / 3 replicas • 3 ClickHouse Keeper nodes • EC2 / EBS • private networking




**CLICKHOUSE ROLE**
Analytical database for LangSmith spans, traces and feedback. Runs OUTSIDE EKS, on dedicated EC2 instances.

**REPLICATION**
One shard with 3 ReplicatedMergeTree replicas across zones. Each keeps its own EBS-backed copy. Any healthy node can serve reads and writes.

**KEEPER QUORUM**
Three Keeper servers maintain shared coordination metadata using Raft; they do NOT store all trace data. Two of three nodes form a majority.

**LANGSMITH CONFIGURATION**
Set clickhouse.external.enabled, host, HTTP/native ports, credentials, TLS and cluster=. Set cluster BEFORE initial migrations so replicated engines are created. LangSmith does not handle read/write routing.

**OPERATIONS**
Back up data to S3 and test restores. Size EC2 and gp3 EBS based on write rate, retention and queries. Monitor insert latency, replica lag, disks and Keeper quorum.

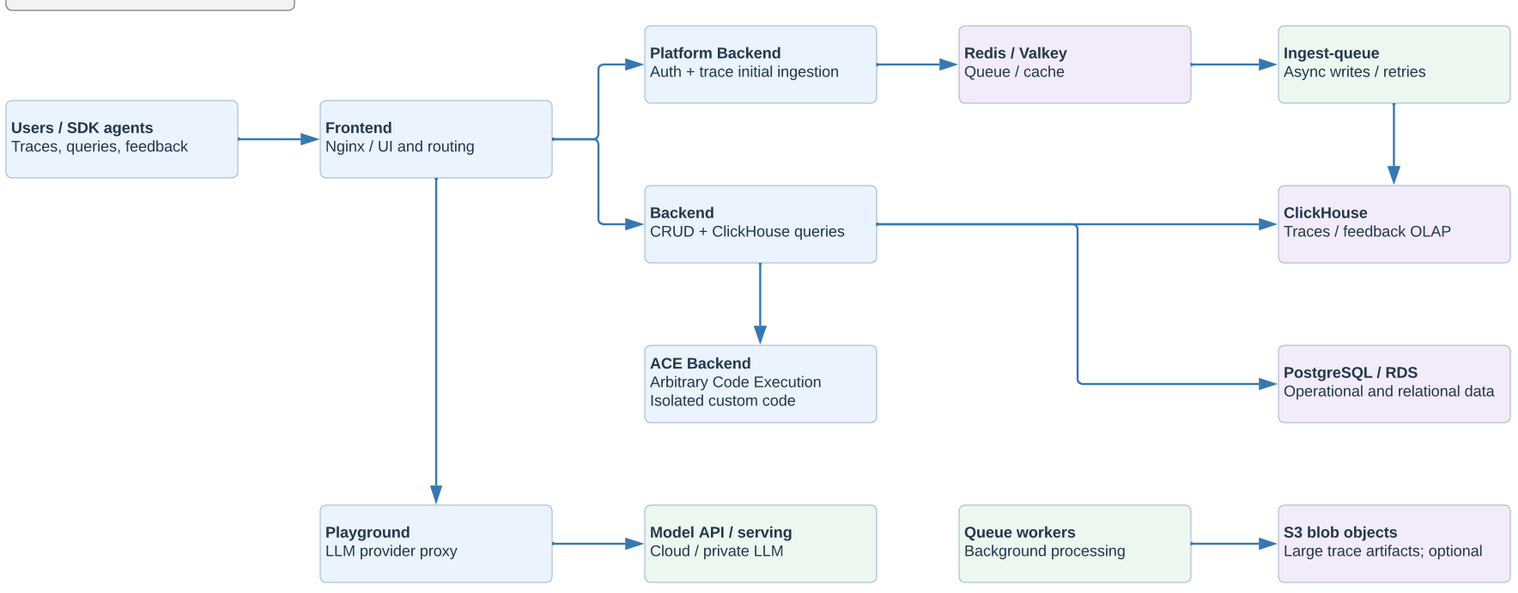
**INTERVIEW TRADEOFF**
Managed ClickHouse is simpler. This is the self-hosted alternative with full operational responsibility.

**Private connectivity**
EKS traffic travels within VPC networking.
Secure with security groups, TLS and credentials.

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## LangSmith Self-Hosted | Read / Write Paths + Operations

Blue = LangSmith API service
Green = workers
Violet = persistent services.




**WRITE PATH — TRACE INGEST**
SDK → Frontend → Platform Backend → Redis → Ingest-queue → ClickHouse. Optimize Platform Backend response latency, Redis pressure, queue backlog, ClickHouse insertion throughput. Queue can separately process background work.

**READ PATH — TRACE QUERY**
UI/API → Frontend → Backend → ClickHouse. Scale Backend pods if request CPU saturates; scale ClickHouse capacity/replicas if queries slow. Independent read and write paths let spikes be managed separately.

**RELIABILITY & SCALING**
Pod HPA, KEDA queue scaling, node autoscaling and multi-AZ spreading handle compute load. External datastores require own HA/backup. S3 is optional but recommended. Data TTL affects ClickHouse sizing.

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## Runtime Architecture and Flows

